

Article

Preparation and Application of Wetland-Plant-Derived Biochar for Tetracycline Antibiotic Adsorption in Water

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Abstract

Every year, a large amount of antibiotics enter aquatic environments globally through discharging of pharmaceutical wastewater and domestic sewage, emissions from agriculture, and livestock, posing a severe threat to ecosystems and human health. Therefore, it is essential to develop efficient adsorption materials for rapid removal of antibiotics in water. In this study, abundant and renewable wetland plants (lotus leaves, *Arundo donax*, and canna lilies) were utilized as raw materials to prepare biochar through slow pyrolysis combined with KOH chemical activation. The prepared biochar was employed to adsorb typical tetracycline (TC) antibiotics (TC-HCl, CTC-HCl, OTC-HCl) from water. The results showed that the optimum biochar (LBC-600 (1:3)) was prepared at a pyrolysis temperature of 600 °C with the mass ratio of KOH to lotus leaf of 1:3. The optimum pH for the adsorption of the three antibiotics were 5, 4, and 3, respectively. The highest adsorption rates reached 93.32%, 81.44%, and 83.76% for TC-HCl, CTC-HCl, and OTC-HCl with 0.6 g/L of biochar, respectively. At an initial antibiotic concentration of 80 mg·L⁻¹, the maximum adsorption capacities achieved 40.17, 27.76, and 24.6 mg·g⁻¹ for TC-HCl, CTC-HCl, and OTC-HCl, respectively. The adsorption process conformed to the pseudo-second-order kinetic and Langmuir isotherm models, indicating that it was a spontaneous endothermic process and primarily involved monolayer chemical adsorption. This study transformed wetland plant waste into adsorbent and applied it for antibiotic removal, providing a valuable resource utilization strategy and technical support for recycling wetland plant residues and antibiotic removal from water environments.

Keywords: wetland plants; biochar; antibiotics; adsorption mechanism



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1. Introduction

Biochar is typically prepared through pyrolysis under oxygen-deficient or low-oxygen conditions. The main types of pyrolysis include slow pyrolysis, microwave-assisted pyrolysis [1], and hydrothermal carbonization [2]. Compared with activated carbon, biochar is relatively inexpensive and possesses strong adsorption capacity. Biochar production is derived from a wide range of sources, mainly including plants, animal manure, and sludge [3]. Its tremendous potential in fields such as water pollution control, soil improvement, and carbon sequestration has been demonstrated.

The total area of wetlands in China is approximately 56.35 million hectares, with 4220 species of wetland plants. Wetland ecosystems generate a large amount of plant residues annually. Without improper treatment, they will accumulate in water or on soil surfaces over a long period, and some of them may decompose under natural conditions. Previous studies demonstrated that plant residues rapidly release nitrogen, phosphorus, and organic matters during their initial decomposition process [4], further deteriorating the environment. Due to their rich organic components such as cellulose, hemicellulose, and lignin, wetland plant residues exhibit multiple properties including high water retention and adsorption capacity. Thus, they have been gradually used to produce some functional materials to achieve their recycling in environmental protection.

As a densely populated country, China has become a major consumer and producer of antibiotics. The commonly used types of antibiotics mainly include sulfonamides, quinolones, macrolides, and tetracyclines [5]. Tetracyclines (TCs) are compounds with strong polarity and good water solubility. Therefore, they are miscible with bases, acids, and polar organic solvents (especially alcohols) but insoluble in saturated hydrocarbons. They are prone to chelation reactions with metal ions, leading to difficulties in their separation and removal from the aquatic environment [6]. Due to the lack of efficient removal methods, approximately 70% of TCs are released into the environment annually through several pathways such as water circulation, soil migration, and bioaccumulation [7]. The present technologies for TC removal mainly include oxidation methods [8], membrane separation techniques [9,10], and adsorption methods [11]. Compared with chemical methods, the adsorption process requires more simple operation and exhibits certain stability in pollutant removal [11]. However, the recovery of adsorbent products requires complex separation and elution procedures. Meanwhile, the adsorption capacity of adsorbents gradually decreases with an increase in the number of cycles [12]. Additionally, the adsorption process is influenced by factors such as temperature, pH, adsorption time, initial concentration, and adsorbent dosage. Thus, exploration of new adsorbents for TC removal is of great significance.

At present, many studies have been focused on plant-derived biochar for antibiotic removal. Chen et al. utilized rice-husk-ash-based biochar to remove antibiotics from water. The results indicated that the removal efficiency of antibiotics was related to the initial concentration of the antibiotic solution, while the adsorption capacity was influenced by the solution's pH value, temperature, and ionic strength, with a maximum adsorption capacity of 8.37 mg/g [13]. Xu et al. employed phosphoric-acid-modified calamus biochar (PBC) for antibiotic adsorption in water [14]. Their findings revealed that PBC contained a significant amount of oxygen-containing functional groups and was capable of adsorbing 325 mg·g⁻¹ of erythromycin (ERY) and 216 mg·g⁻¹ of sulfamethoxazole (SMX), which was 10 times higher than that of unmodified biochar and commercial activated carbon. The maximum antibiotic adsorption performance was observed when PBC was prepared with an impregnation ratio of 3, an impregnation time of 6 h, and a carbonization temperature of 536 °C. Phisit Thairattananon et al. used FeCl₃-pyrolyzed magnetic-biochar-impregnated watermelon rind to remove tetracycline from water [15]. The study found that the material synthesized at 900 °C possessed more adsorption sites, and the optimal adsorption capacity was achieved at pH = 3, with an adsorption capacity of 77.6 mg·g⁻¹.

Biochar exhibits significant potential in pollutant removal from water environments. Previous studies mainly focused on biochar production from agricultural waste and sludge. This study aims to utilize abundant wetland plant residues for biochar production and employ the prepared biochar for common antibiotic (TC) removal in a water environment via an adsorption process. The results of this study can provide a valuable resource

utilization strategy and technical support for recycling wetland plant residues and antibiotic removal from water environments.

2. Materials and Methods

2.1. Experimental Materials

The wetland plants used in this study, including canna (*Canna indica* Linn.), lotus leaf (lotus leaf), and giant reed (*Arundo donax* Linn.), were collected from Sanyang Wetland in Wenzhou City, Zhejiang Province. After collection, the plants were rinsed with distilled water to remove surface impurities and then placed in a forced-air drying oven for 24 h of drying at 80 °C.

The antibiotics selected for this experiment were tetracycline hydrochloride (TC-HCl), chlortetracycline hydrochloride (CTC-HCl), and oxytetracycline hydrochloride (OTC-HCl), all of which were USP grade and were purchased from Macklin Biochemical Technology Co., Ltd., Shanghai, China.

2.2. Preparation of Biochar

2.2.1. Labeling of Biochar

To facilitate the differentiation and efficient management of biochar samples prepared from different precursors and modified using various approaches during the experimental process, as well as to enable systematic comparison and summarization, this study adopted an abbreviated naming system for all biochar samples. The raw biochars were categorized based on the first letter of the English names of their precursors, with “BC” serving as the base name for biochars. Specifically, biochars derived from different precursors were distinguished by the initial letters of their Latin or English names. For instance, biochar produced from lotus leaves was designated as LBC, canna-based biochar as CBC, and biochar from giant reed as ABC. For biochars generated at different pyrolysis temperatures, the temperature was appended to the base name. For example, lotus leaf biochar produced at 500 °C was named LBC-500. Regarding biochars modified with different KOH ratios, the ratio was indicated after the biochar name. For instance, lotus leaf biochar modified at 500 °C with a mass ratio of 1:3 (lotus leaf to KOH) was named LBC-500 (1:3).

2.2.2. Preparation of Primitive Biochar

The dried lotus leaves were cut into small segments approximately 3 cm in length. These segments were then ground using a pulverizer for 2 min each time. The ground lotus leaf powder was mixed with distilled water in a 55 mL PP centrifuge tube to form a homogeneous suspension. Subsequently, the suspension was filtered through a 100-mesh sieve. The filtrate was placed in a low-speed centrifuge and subjected to centrifugation for 3 min at a speed of 4200 rpm. The solid powder retained at the bottom was then transferred to an 80 °C oven for 24 h of drying. Thirty grams of the dried sample were evenly placed in a crucible, which was wrapped externally with aluminum foil. The wrapped crucible was placed in a muffle furnace, and the temperature was set to 300 °C. Pyrolysis was carried out for 3 h at a heating rate of 10 °C·min^{−1}. After pyrolysis, the crucible was removed and cooled to room temperature. The solid material obtained at this stage was the raw biochar. The biochar was washed with distilled water and filtered using a 0.45 µm filter paper, with the washing and filtration process repeated three times. The washed biochar was then placed in a 120 °C oven and dried overnight. The dried sample was ground using an agate mortar, passed through a 50-mesh sieve, and weighed. The preparation methods for biochars from giant reed and canna were the same as described above.

2.2.3. Preparation of KOH-Modified Biochar with Different Proportions

Thirty grams of KOH powder was taken and dissolved in 400 mL of deionized water. Subsequently, three types of plant powders were respectively mixed with the KOH solution in a 55 mL PP centrifuge tube at mass ratios of 3:1, 2:1, 1:1, 1:2, and 1:3. The resulting suspensions were stirred at a speed of 800 rpm. Subsequently, the mixtures were placed in a centrifuge and centrifuged at 4200 rpm for 3 min. After centrifugation, the solid materials in the centrifuge tubes were retained and transferred to an ultra-low-temperature freezer, where they were frozen at $-80\text{ }^{\circ}\text{C}$ for 12 h. Upon completion of freezing, the materials were placed in a vacuum dryer for vacuum drying at 450.0 Pa and $-37.0\text{ }^{\circ}\text{C}$. The vacuum-dried materials were then placed in crucibles and subjected to pyrolysis by following the steps outlined in Section 2.2.2.

2.2.4. Preparation of Biochar at Different Pyrolysis Temperatures

The untreated raw materials and the precursors that had undergone alkaline impregnation treatment were placed in a muffle furnace for thermal treatment according to the method described in Section 2.2.2. All experimental conditions remained unchanged except for the temperature. According to previous studies, biochar prepared at different temperatures usually exhibits significant differences in chemical composition, pore structure, specific surface area, functional groups, and stability. Temperatures of 300 to 600 $^{\circ}\text{C}$ can better reflect the gradual changes in these properties. This temperature range can also allow for relatively controllable energy consumption in both laboratory and industrial preparations, as well as offers a balance in experimental research and practical production. At present, 300 to 600 $^{\circ}\text{C}$ is commonly used in biochar preparation research with a reasonable span (one gradient every 100 $^{\circ}\text{C}$). Thus, 300 $^{\circ}\text{C}$, 400 $^{\circ}\text{C}$, 500 $^{\circ}\text{C}$, and 600 $^{\circ}\text{C}$ were selected for the pyrolysis experiments, where the heating rate was set at $10\text{ }^{\circ}\text{C min}^{-1}$.

The flow chart for biochar preparation is shown in Figure 1.

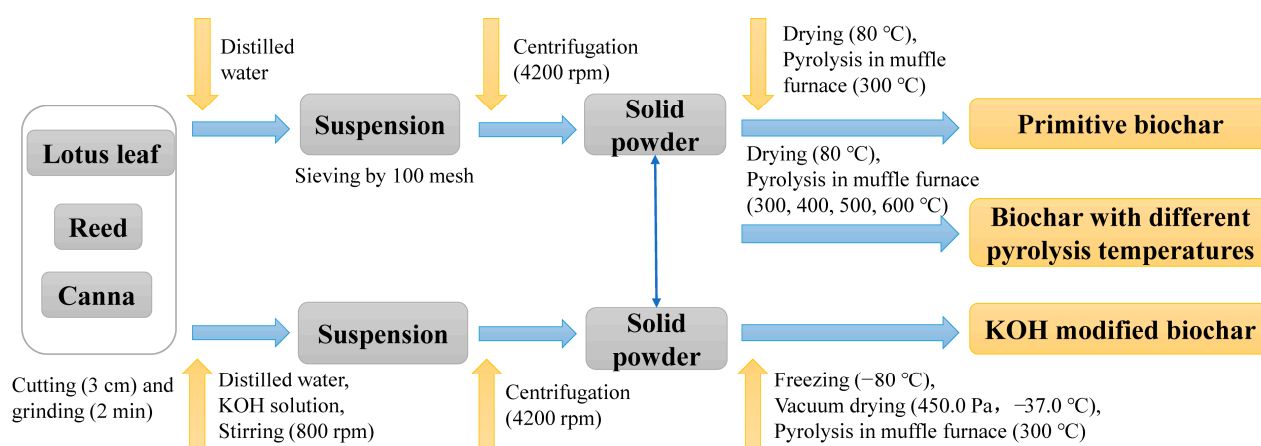


Figure 1. Flow chart for the preparation of biochar.

2.3. Analysis and Testing Methods

2.3.1. Antibiotic Adsorption Experiment

Single-factor experiments (investigating factors such as pH, biochar dosage, and initial concentration of adsorbate), experiments involving coexisting ions, and cyclic adsorption experiments were conducted to explore the effects of various adsorption-influencing factors on the adsorption performance of biochar and to quantitatively assess the stability of biochar. Additionally, adsorption kinetics and adsorption thermodynamics experiments were set up to analyze the relevant adsorption behaviors of biochar.

The antibiotic solutions were prepared with a concentration of $20 \text{ mg}\cdot\text{L}^{-1}$. The pH values of TC-HCl, CTC-HCl, and OTC-HCl solutions were measured to be 2.38, 2.80, and 2.86, respectively. The pH of the antibiotic solutions was adjusted to 3 using 0.1 M hydrochloric acid and sodium hydroxide solutions. The adsorption experiments were conducted at a temperature of 25°C . In 55 mL centrifuge tubes, 30 mg of biochar was added, followed by 50 mL of each of the three antibiotic solutions with a concentration of $20 \text{ mg}\cdot\text{L}^{-1}$.

2.3.2. Detection Methods for Antibiotics

An ultraviolet–visible spectrophotometer was employed to detect antibiotics. The detection wavelengths were set as follows: 357 nm for TC-HCl, 367 nm for CTC-HCl, and 354 nm for OTC-HCl. The tetracycline solution in the centrifuge tube was filtered through a $0.45 \mu\text{m}$ filter membrane. Subsequently, the adsorbed sample solution was diluted 500-fold, and the relevant absorbance was measured. The calculation formula for the adsorption efficiency of antibiotics by biochar is provided in Equation (1).

$$R = \frac{(C_0 - C_e)}{C_0} \cdot 100\% \quad (1)$$

In the formula: R represents the removal rate (%) of tetracycline by biochar, C_0 is the initial concentration ($\text{mg}\cdot\text{L}^{-1}$), and C_e is the concentration after adsorption ($\text{mg}\cdot\text{L}^{-1}$).

2.3.3. Microscopic Testing Methods

The surface morphology of biochar was analyzed using a scanning electron microscope (Gemini SEM 360, Carl Zeiss AG, Baden-Württemberg, Germany). Elemental analysis of the biochar was conducted using an elemental analyzer (Flash Smart, Thermo Fisher Scientific Inc., Waltham, MA, USA). The surface functional groups of the biochar were examined using a Fourier-transform infrared spectrometer (TENSOR 27, Bruker Corporation, Würzburg, Germany). The surface elements, chemical states, and molecular structures of the target biochar were investigated using an X-ray photoelectron spectrometer (ThermoFisher Nexsa, Thermo Fisher Scientific Inc., MA, USA). The specific surface area and pore structure of the biochar were measured using a specific surface area and pore size analyzer (BELSORP MAX, MicrotracBEL Corp., Osaka, Japan).

2.3.4. Common Salts and Recycling Experiments

To investigate the influence of common salts on the adsorption performance of biochar, cations (Na^+ , Ca^{2+} , Mg^{2+}) and anions (Cl^- , SO_4^{2-} , NO_3^-) present in water were selected for the experiments. Each of these ions was added separately to three types of antibiotic solution (with a concentration of $20 \text{ mg}\cdot\text{L}^{-1}$). The concentrations of the corresponding cations and anions were adjusted to 0, 10, 20, and $30 \text{ mmol}\cdot\text{L}^{-1}$, respectively. Then, 30 mg of LBC-600 (1:3) was added to each solution. The solutions were subjected to a 24 h oscillation reaction at 180 rpm in a constant-temperature shaking incubator at 25°C .

To explore the stability of the biochar, cyclic adsorption experiments were set up. The adsorption of three antibiotics by LBC-600 (1:3) was carried out under the experimental conditions described in Section 2.3.1. After adsorption, the mixed solution was separated into solid and liquid phases using a suction filtration pump and a $0.45 \mu\text{m}$ aqueous microporous filter membrane (the suction filtration apparatus was dried in an 80°C oven before use). The antibiotic concentration in the filtrate was measured using a spectrophotometer. The separated biochar was then subjected to a 24 h oscillation desorption process using a $0.8 \text{ mol}\cdot\text{L}^{-1}$ NaOH desorbent (at 180 rpm and 25°C). The desorbed biochar was washed three times with deionized water and dried at 80°C . After drying, the biochar was reused

for cyclic adsorption experiments under the same conditions with the same dosage as in the adsorption experiments. The cyclic experiments were conducted six times. Each experiment was carried out in triplicate.

2.3.5. Adsorption Kinetics

To investigate the adsorption behavior of LBC-600 (1:3) towards three antibiotics (TC-HCl, CTC-HCl, OTC-HCl) and to identify the dominant mechanism in the adsorption process, relevant adsorption kinetic experiments were conducted. The adsorption times were set at 0, 0.05, 0.17, 0.33, 0.5, 1, 2, 3.33, 4, 5, 6, 8.33, 13.33, 16.67, 20, and 24 h, respectively. The concentrations of all three antibiotics were $20 \text{ mg}\cdot\text{L}^{-1}$. A dosage of 30 mg of LBC-600 (1:3) was added to the solutions, which were then oscillated at a speed of 180 rpm at 25°C . Triplicate parallel experiments were carried out for each adsorption time.

The experimental data were fitted using the pseudo-first-order (2) and pseudo-second-order (3) kinetic models and the intraparticle diffusion model (4), with their equations presented as follows:

$$Q_t = Q_e \left(1 - e^{-k_1 t}\right) \quad (2)$$

$$Q_t = \frac{k_2 Q_e^2 t}{1 + k_2 Q_e t} \quad (3)$$

$$Q_t = k_p t^{0.5} + C \quad (4)$$

where Q_t represents the adsorption capacity at time t ($\text{mg}\cdot\text{g}^{-1}$); Q_e ; refers to the adsorption capacity per unit mass of the adsorbent when adsorption equilibrium is reached ($\text{mg}\cdot\text{g}^{-1}$); t is the adsorption time (h); k_1 , k_2 , and k_p ; are the rate constants for the pseudo-first-order, pseudo-second-order, and intraparticle diffusion models, respectively (h^{-1} , $\text{g}\cdot\text{mg}^{-1}\cdot\text{h}^{-1}$, and $\text{mg}\cdot\text{g}^{-1}\cdot\text{h}^{-0.5}$); C is a constant related to the thickness of the adsorption boundary layer.

2.3.6. Adsorption Isotherm

The concentrations of the three antibiotics were set at 3, 5, 10, 15, 20, 25, 30, 40, 50, 55, 60, 70, 80 $\text{mg}\cdot\text{L}^{-1}$. A dosage of 30 mg of LBC-600 (1:3) was added, and the mixtures were oscillated at a speed of 180 rpm for 24 h. Triplicate parallel experiments were carried out for each antibiotic concentration.

The relevant adsorption isotherms were fitted using the Langmuir and Freundlich models, with the corresponding Equations shown as (5) and (6), respectively.

The Langmuir adsorption isotherm model:

$$Q_e = \frac{Q_m K_a C_e}{1 + K_a C_e} \quad (5)$$

The Freundlich adsorption isotherm model:

$$Q_e = K_f C_e^{\frac{1}{n}} \quad (6)$$

where Q_e represents the adsorption capacity of biochar for the relevant adsorbate ($\text{mg}\cdot\text{g}^{-1}$); Q_m is the maximum adsorption capacity of the adsorbent ($\text{mg}\cdot\text{g}^{-1}$); C_e denotes the concentration of the antibiotic solution at adsorption equilibrium ($\text{mg}\cdot\text{L}^{-1}$); K_a is the Langmuir constant ($\text{L}\cdot\text{mg}^{-1}$); K_f is the Freundlich affinity coefficient related to the adsorption capacity; and $1/n$ is a constant associated with the adsorption energy.

3. Results and Discussion

3.1. Optimization of Preparation Conditions for Biochar

3.1.1. Impact of the KOH Proportion on the Adsorption Effectiveness of Modified Biochar

The adsorption efficiencies of three antibiotics (TC-HCl, CTC-HCl, and OTC-HCl) by biochars modified with KOH at different mass ratios are presented in Figure 2. As the proportion of KOH added increased, the adsorption efficiencies of the biochars for all three antibiotics exhibited a growing trend. Thus, it can be concluded that there was a positive correlation between the degree of KOH modification of the biochars and their adsorption capacities.

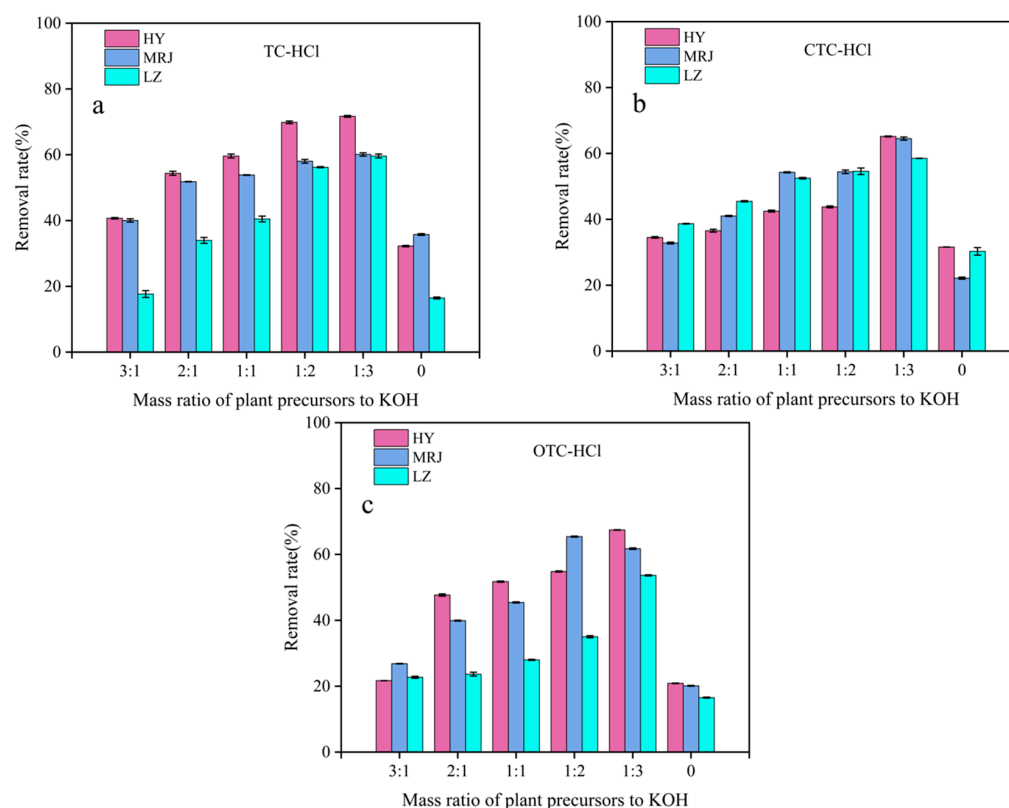


Figure 2. Adsorption efficiency of TC-HCl (a), CTC-HCl (b), OTC-HCl (c) by preparing biochar with different KOH mass ratios (pH = 3.0, V = 50 mL, T = 25 °C, C₀ = 20 mg·L^{−1}, m = 30 mg, t = 24 h). HY: lotus leaf biochar; MRJ: plantain biochar; LZ: reed bamboo biochar.

The results from Figure 2a–c indicate that lotus leaf biochar demonstrated the best adsorption efficiency in removing the three antibiotics, namely tetracycline (TC-HCl), chlortetracycline (CTC-HCl), and oxytetracycline (OTC-HCl), under different KOH modification ratios. In particular, when the KOH ratio was 1:3 (mass ratio of biochar precursor to KOH), its adsorption efficiency significantly improved. The adsorption efficiencies for TC-HCl, CTC-HCl, and OTC-HCl reached 71.68%, 65.16%, and 67.43%, respectively, which were the highest values among the adsorption efficiencies of the three biochars at this ratio. The yields of LBC-500 and LBC-600 before the addition of KOH were 317 and 304 mg g^{−1}, respectively. After 1:3 of KOH was added, their yields were changed to 301 and 284 mg g^{−1}.

Therefore, subsequent research will focus on exploring the optimal application conditions of this ratio (a mass ratio of 1:3 between the biochar precursor and KOH) in the modification of different plant precursors. The aim is to further optimize the performance of biochars and achieve more efficient removal of antibiotics.

3.1.2. Influence of Pyrolysis Temperature on the Adsorption Effectiveness of Modified Biochar

Figure 3 illustrates the adsorption effects of biochars prepared at different pyrolysis temperatures on three types of antibiotics. Consistent with the experimental results presented in Section 3.1.1, an increase in pyrolysis temperature significantly enhanced the adsorption efficiency of the biochars for all three antibiotics. As the temperature rose, the adsorption efficiency of lotus leaf biochar gradually increased. In contrast, although the adsorption efficiencies of canna biochar and arundo biochar also improved with increasing temperature, they remained consistently lower than those of lotus leaf biochar, with relatively smaller increments, failing to surpass the adsorption performance of lotus leaf biochar for antibiotics.

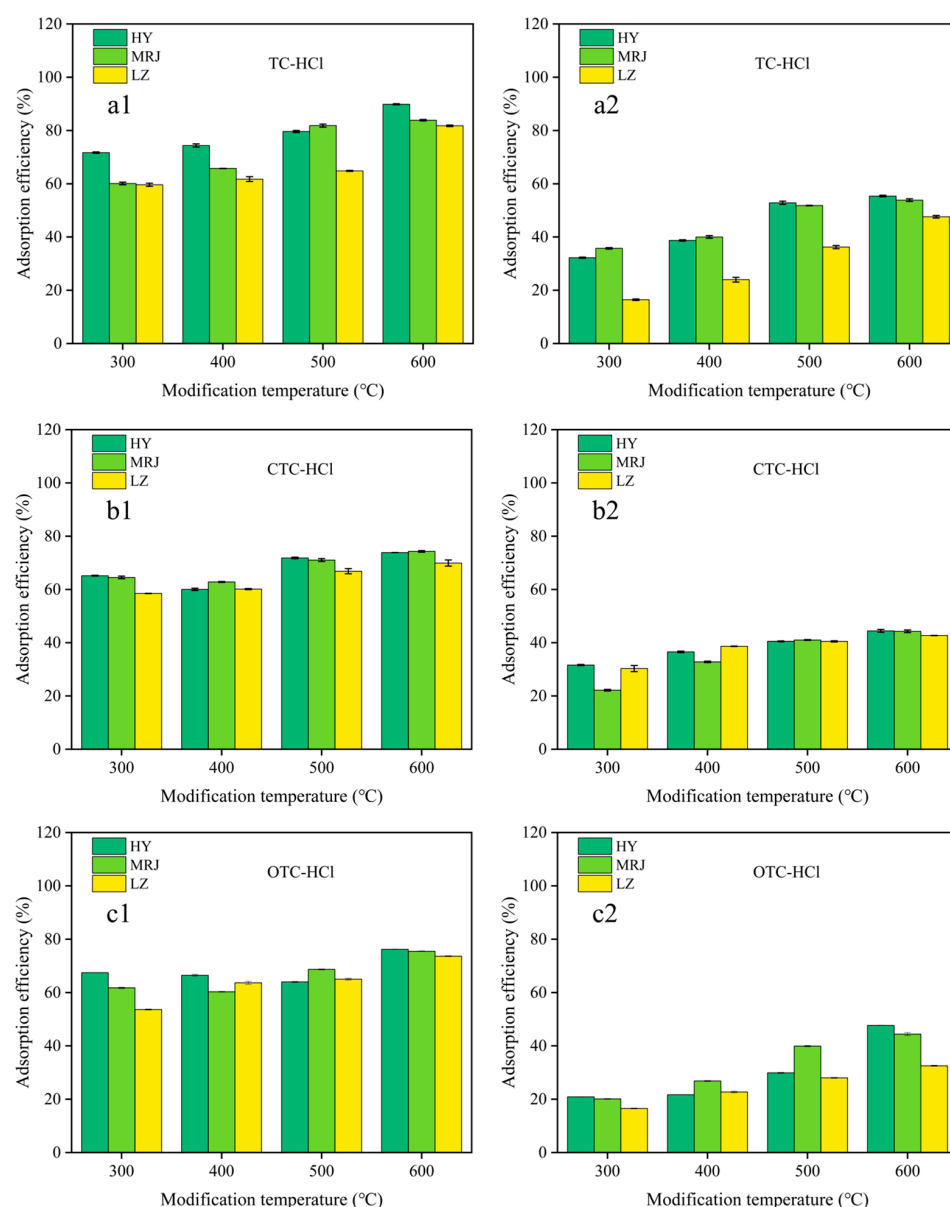


Figure 3. Effect of different temperature-modified biochars on adsorption of three kinds of antibiotics (The ratio of biochar precursors to KOH was 1:3, $T = 25\text{ }^{\circ}\text{C}$, $m = 30\text{ mg}$, $V = 50\text{ mL}$, $C_0 = 50\text{ mg}\cdot\text{L}^{-1}$, $\text{pH} = 3.0$). (a1–c1): Pyrolysis temperature modification when mass ratio of biochar precursors to KOH is 1:3; (a2–c2): pyrolysis temperature modification of original biochar). HY: lotus leaf biochar; MRJ: plantain biochar; LZ: reed bamboo biochar.

The adsorption efficiency of biochars for antibiotics was positively correlated with the pyrolysis temperature, indicating that the influence of pyrolysis temperature on the antibiotic adsorption performance of biochars was continuous and gradual. A possible explanation for this phenomenon is that the surface area of biochars increases with temperature, thereby enhancing their adsorption capacity for antibiotics [16].

Meanwhile, it can be observed from the figure that the adsorption efficiencies of the twice-modified biochars for the three antibiotics were roughly similar at 600 °C. This result may be related to the similarities in the chemical properties and molecular structures of the three antibiotics. Furthermore, this outcome also suggests that the modification effect of pyrolysis temperature on biochars reached a certain equilibrium, leading to a stabilization of adsorption efficiency. Therefore, it can be concluded that 600 °C is the optimal temperature for the pyrolysis modification of biochars in experiments, as it allows biochars to achieve a good balance between antibiotic adsorption efficiency and surface area.

3.2. Microscopic Characterization of Biochar Properties

3.2.1. SEM

The scanning electron microscopy–energy dispersive spectroscopy (SEM-EDS) images of the biochars are shown in Figure 4, and the elemental mass percentages at the scanned points are presented in Table 1. Observing the biochar (a) produced through pyrolysis at 500 °C, its surface characteristics exhibit a relatively rough texture with distinct granularity. The particles are spaced apart with a loose arrangement, indicating low structural density overall. Despite the rough substrate, the particles themselves appear relatively smooth, creating a contrast. When the pyrolysis temperature was increased to 600 °C, certain changes were observed in the microstructure of the resulting biochar (c). On the whole, its surface morphology resembles that of biochar (a) produced at 500 °C, still displaying a rough and fragmented texture. However, the particle distribution in biochar (c) is more concentrated compared to biochar (a), with reduced particle spacing and locally denser arrangements. Additionally, the external morphology of biochar (c) appears sharper and more irregular. The differences in surface morphology between biochar (a) and (c) may be attributed to the higher temperature, which facilitated further cracking of organic compounds in the biomass, releasing more volatile components (such as CO, CO₂, and water vapor). This, in turn, influenced the arrangement state and particle morphology on the material's surface.

The surface morphology images of the KOH-modified biochars are shown in Figure 4b,d. Compared with biochar a, under a pyrolysis environment of 500 °C, the addition of KOH significantly increased the pore structure of the biochar. When comparing Figure 4b with Figure 4d, under the same mass of KOH modification, an increase in temperature led to a substantial enhancement of the pore structure in the biochar compared to Figure 4b. Specifically, at a pyrolysis temperature of 500 °C, the surface pore structure of biochar a was still in a relatively preliminary stage. After KOH modification, sporadic micropores were visible in some areas of the surface, with a small number of unevenly distributed pores and small pore sizes (mainly micropores). The pores were scattered and did not form a well-connected pore network. This may be attributed to the lower temperature, which limited the in-depth reaction between KOH and the biochar, failing to fully activate the internal skeletal structure. As a result, a large number of potential pores had not yet emerged. At 600 °C, the number of pores in biochar d significantly increased, and their distribution became more uniform. Larger-scale pores appeared on the surface, giving it a more three-dimensional appearance. This is consistent with the viewpoint of Zhang et al. who believed that an increase in temperature would lead to an increase in the roughness of the biochar surface [17].

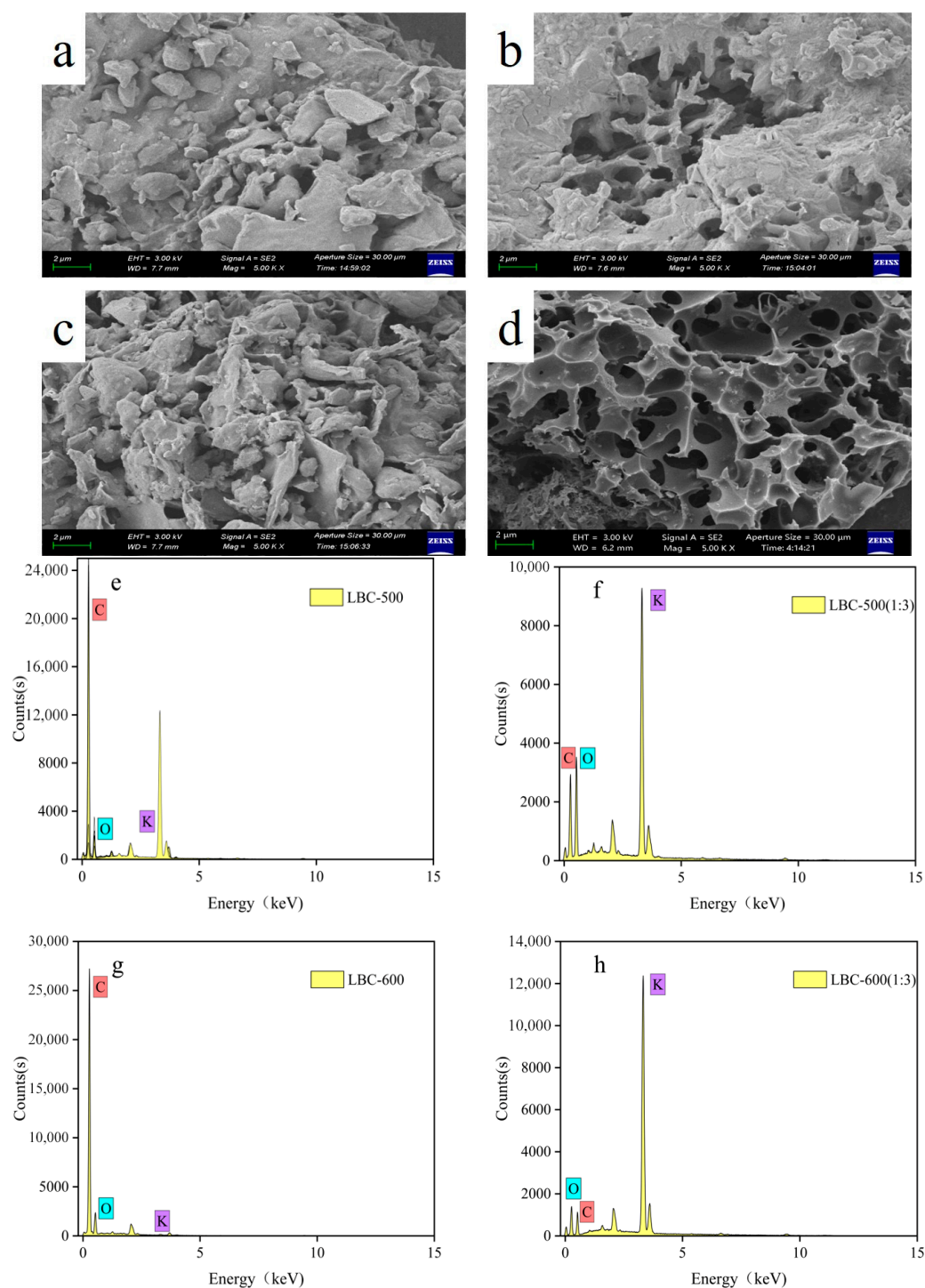


Figure 4. SEM-EDS diagram of biochar SEM: (a): LBC-500, (b): LBC-500 (1:3), (c): LBC-600, (d): LBC-600 (1:3) EDS, (e): LBC-500, (f): LBC-500 (1:3), (g): LBC-600, (h): LBC-600 (1:3).

Table 1. Mass specific gravity of elements at the energy spectrum scanning point.

Name of Biochar	C (wt %)	N (wt %)	O (wt %)	K (wt %)	Cr (wt %)
LBC-500	83.04	0.03	16.45	0.44	0.03
LBC-500 (1:3)	13.60	0.77	32.29	46.14	0.21
LBC-600	79.82	3.29	16.69	0.19	0.01
LBC-600 (1:3)	6.78	1.65	13.85	77.44	0.28

Energy dispersive spectroscopy (EDS) scanning provides information on the elemental distribution within a single region on the biochar surface. This method can be utilized to qualitatively characterize the local differences in surface modification or elemental enrichment of biochar materials. Compared with regions e and g, regions f and h exhibited exceptionally high K and O contents. From this, it can be inferred that elements from KOH were successfully introduced onto the biochar surface.

3.2.2. XPS

The surface chemical compositions of four types of biochar materials (LBC-500, LBC-500 (1:3), LBC-600, and LBC-600 (1:3)) were analyzed using X-ray photoelectron spectroscopy (XPS) technology. The survey spectra of the four biochars are presented in Figure 5. From the XPS survey spectra, characteristic peaks corresponding to carbon and oxygen elements were observed in all samples, indicating that carbon and oxygen are the primary constituent elements of these four biochar materials. Additionally, a characteristic peak for nitrogen (N1s) was detected near 400 eV in the spectra of all samples, albeit at relatively low contents. Quantitative analysis based on atomic percentages revealed that the nitrogen contents were 3.83%, 0.95%, 5.80%, and 0.56%, respectively, which is consistent with the results obtained from elemental analysis.

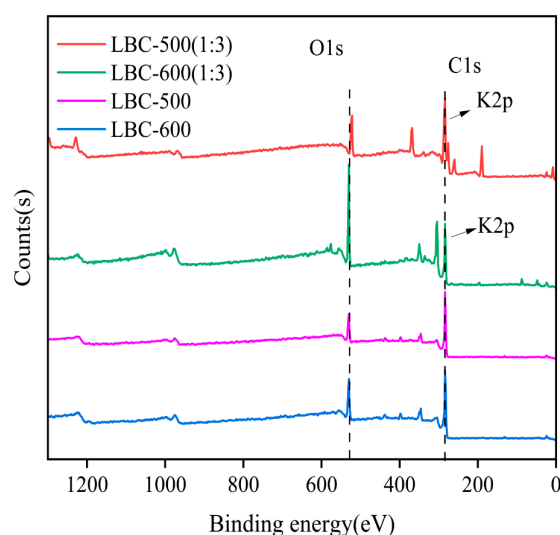


Figure 5. Full spectrum of four biochar species.

The proportion of chemical states of relevant elements could be inferred from the magnitudes of the peak areas [18]. It is noteworthy that characteristic peaks corresponding to potassium were not detected in the biochar samples that had not undergone KOH modification treatment. However, in the biochar materials modified with KOH (namely, LBC-500 (1:3) and LBC-600 (1:3)), distinct characteristic peaks were observed at 294.08 eV and 296.06 eV, respectively, corresponding to the K2p (3/2) and K2p (1/2) orbitals of the potassium (K) element. The emergence of signals from the K2p (3/2) and K2p (1/2) orbitals provides evidence that KOH has been successfully introduced onto the surface of the biochars and effective bonding has been achieved among the biochars.

The C1s spectra of the four types of biochars (Figure 6) revealed that, compared to the unmodified biochars, the KOH-modified biochars exhibited a significant increase in both peak intensity and the variety of peaks. An additional O-C=O peak (20.40%) at 288.5 eV was observed in biochar b. Similarly, biochar d also showed a new O-C=O peak (5.04%) at the same binding energy of 288.5 eV. This indicates that the introduction of KOH played a significant role in the formation of oxygen-containing functional groups on the biochar

surfaces. For biochars b and d, C=O peaks were formed at 289 eV (b: 16.55%, d: 4.99%, respectively). The difference in peak intensities can be attributed to that an increase in temperature leads to further graphitization of the biochar material, reducing the number of surface functional groups [19].

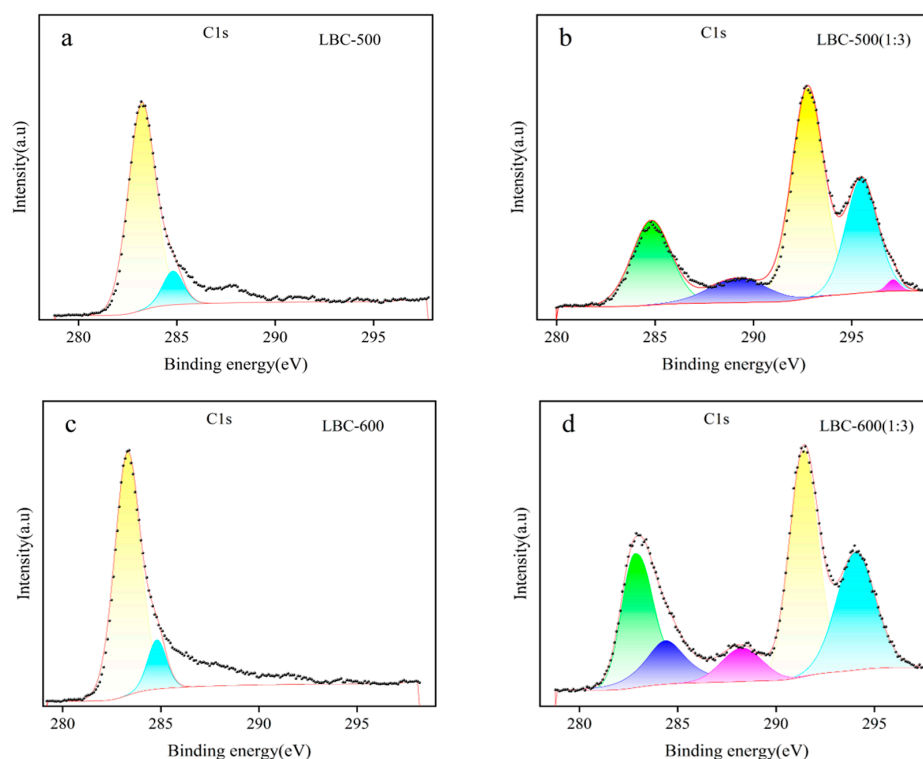


Figure 6. C1s spectra of four target biochars before adsorption (a): LBC-500, (b): LBC-500 (1:3), (c): LBC-600, (d): LBC-600 (1:3).

K2p characterization analyses were conducted on LBC-500 (1:3) and LBC-600 (1:3), and the results are presented in Figure 7. It can be observed from the figure that, in comparison with biochars a, c and b, d, two K2p characteristic peaks were generated after KOH modification. The binding energy positions of these peaks were as follows: for biochar b, they were at 292.87 eV (61.98%) and 295.58 eV (34.60%); for biochar d, they were at 291.81 eV (58.52%) and 294.45 eV (41.48%). The binding energy differences between each pair of characteristic peaks in the figure were similar ($\Delta E = 2.71$; $\Delta E = 2.64$). Since the double peaks in the K2p photoelectron spectrum are typically formed through spin–orbit splitting, the gap in binding energy (the spin–orbit splitting value) between the two peaks is usually fixed [20].

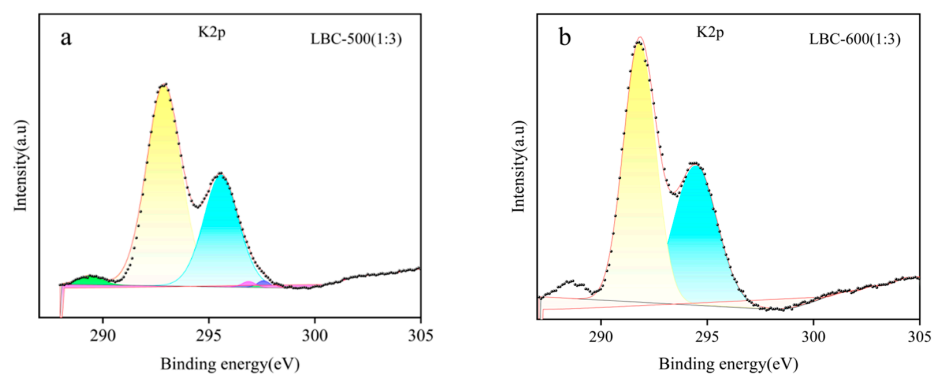


Figure 7. K2p spectra of LBC-500 (1:3) and LBC-600 (1:3). (a): LBC-500 (1:3), (b): LBC-600 (1:3).

3.2.3. FTIR

A more intuitive analysis of the characteristics of relevant chemical bonds could be achieved through Fourier transform infrared (FTIR) spectroscopy. The relevant FTIR images are shown in Figure 8. From the images, it can be observed that all four types of biochars exhibited weak absorption peaks near 873 and 875 cm^{-1} , which may be attributed to the out-of-plane bending vibration of aromatic ring C-H bonds. These peaks result from para-substitution of hydrogen atoms on the aromatic rings in terms of their positions and quantities. The weak peaks are likely the superposition of out-of-plane bending vibrations of aromatic ring C-H bonds with various substitution patterns. Due to the low content of the corresponding structures, these peaks may overlap, leading to the observation of only a single weak peak [21]. Two characteristic peaks were detected at 1405.85 cm^{-1} and 1413.10 cm^{-1} in the two KOH-modified biochars, LBC-500 (1:3) and LBC-600 (1:3), respectively. These peaks originated from the carboxyl groups (-COOH) of carboxylate salts. Studies have shown that COO^- , as a multi-electron conjugated system, exhibits strong absorption in the asymmetric stretching vibration (1560–1610 cm^{-1}) and symmetric stretching vibration (1360–1440 cm^{-1}) regions due to the vibrational coupling of the two C=O bonds. The peaks observed in the figure correspond to the symmetric stretching vibration of carboxyl groups, with their intensities being weaker than those of the asymmetric stretching vibrations and two peaks appearing. This is consistent with the conclusions of Lea Nolte et al. [22]. The vibration peak observed at 1619.12 cm^{-1} in LBC-500 (1:3) is attributed to the C=C bond in aromatic rings, while the characteristic peak observed at 1631.68 cm^{-1} in LBC-600 (1:3) can be ascribed to the -OH bond. The presence of -COOH indicates that the number of oxygen-containing functional groups on the biochar surface significantly increased after KOH modification. Except for LBC-500, the other three types of biochars all exhibited a broad characteristic peak near 3400 cm^{-1} . The formation of this broad peak is related to hydrogen bonding. Data indicate that -OH groups typically appear in the range of 3340–3435 cm^{-1} , so it can be inferred that the characteristic peak observed here is due to -OH groups. After the biochars were modified at high temperatures and with KOH, their specific surface areas increased. Upon exposure to moisture in the air, water molecules interacted strongly with the hydrogen bonds of hydroxyl groups on the biochar surface, resulting in a more pronounced broad -OH peak [23].

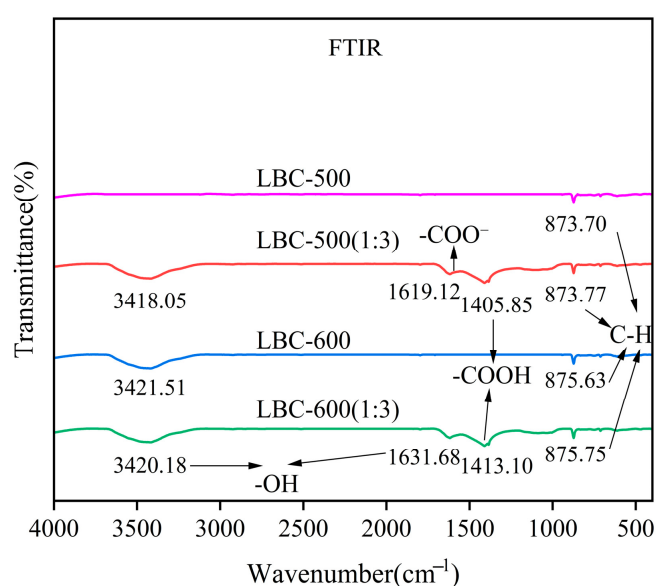


Figure 8. FTIR analysis of biochar.

The presence of oxygen-containing functional groups, such as -COOH , C=O , and -OH , enables the formation of hydrogen bonds with water molecules, significantly enhancing the hydrophilicity of biochars. Among these groups, -OH and -COOH carry positive charges in acidic environments. The existence of these oxygen-containing functional groups facilitates the effective adsorption of antibiotics through interactions involving π - π bonds and hydrogen bonding [24].

3.2.4. XRD

Figure 9 displays the XRD patterns of the four target biochars. Due to the periodic arrangement of atoms, multiple distinct peaks were observed in the XRD patterns of all four biochars. It was proposed that high temperatures can promote the growth of crystal grains, resulting in sharper peaks in the XRD patterns. The results presented in Figure 9a are consistent with this viewpoint [25]. Comparison with PDF cards revealed that LBC-600 matched the calcite phase (PDF#86-0174), whereas for LBC-500, the lower temperature was insufficient to convert CaCO_3 into the stable calcite phase, leading to the formation of the metastable aragonite phase instead (PDF#72-1937). The generation of CaCO_3 originated from the presence of small amounts of alkaline earth metals such as Ca and Mg in the plant residues. During the pyrolysis process, these alkaline earth metals were transformed into minerals existing in the forms of oxides, carbonates, and a small amount of phosphates. When effective combination between Ca^{2+} and CO_3^{2-} occurred during this process, the phases corresponding to LBC-500 and LBC-600 in the XRD patterns were identified as CaCO_3 [26]. The surface of CaCO_3 crystals can serve as adsorption sites, facilitating coordination, ion exchange, and electrostatic adsorption interactions with antibiotic molecules.

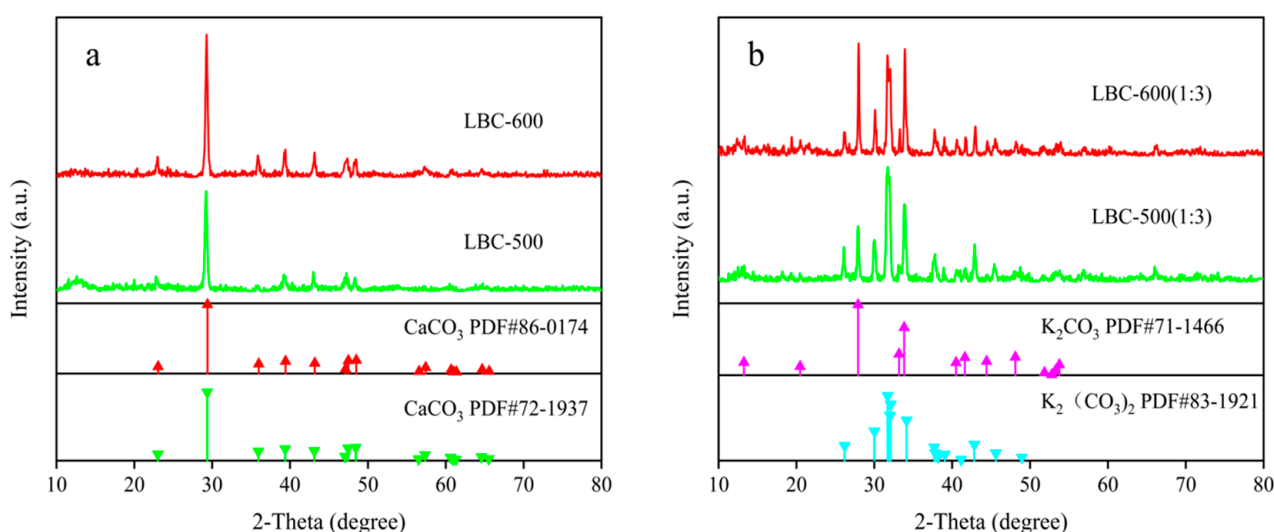


Figure 9. XRD patterns of four kinds of carbon (a): LBC-500 and LBC-600, (b): LBC-500 (1:3) and LBC-600 (1:3).

Figure 9b detected the presence of two mineral phases, K_2CO_3 and $\text{K}_2\text{Ca}(\text{CO}_3)_2$, in both types of biochars. Unlike Figure 9a, the PDF cards corresponding to the two biochars in Figure 9b were consistent. For LBC-600 (1:3), particularly within the 20–40° range, the XRD curve exhibited multiple sharp peaks with high intensity, indicating a high degree of crystallinity in this biochar. In contrast, LBC-500 (1:3) displayed lower peak intensities, with some peaks being broader. According to the Scherrer equation, this suggests that LBC-500 (1:3) had lower crystallinity, smaller crystal grain sizes, and contained more amorphous components. In Figure 9b, K_2CO_3 and CaCO_3 existed as independent carbonate forms

in both biochars, but the content of K_2CO_3 was significantly higher than that of $CaCO_3$. The formation of $K_2Ca(CO_3)_2$ might be attributed to the decomposition of K from KOH and Ca from the plant material at high temperatures, generating K_2O and CaO . In the closed pyrolysis system of the muffle furnace, CO_2 had a long residence time and reacted with these two oxides to initially form K_2CO_3 and $CaCO_3$. At 500 °C, the production of $K_2Ca(CO_3)_2$ was relatively low, and its crystal phase was unstable, which was manifested as broader peaks in the XRD pattern. At 600 °C, with a further increase in temperature, the formed $K_2Ca(CO_3)_2$ had a more stable crystal structure, resulting in sharper XRD peaks and higher crystallinity [27]. Compared to the two biochars in Figure 9a, the KOH-modified biochars provided significantly more adsorption sites than the unmodified biochars [28].

3.2.5. BET

Studies have indicated that the specific surface area of biochar typically increases with rising pyrolysis temperature. Measuring the specific surface area of biochar can provide a more intuitive reference for understanding the adsorption mechanisms of antibiotics onto biochar. Figure 10 presents the nitrogen adsorption–desorption isotherms and pore volume–pore size distribution diagrams for four types of biochars. According to the classification method of the International Union of Pure and Applied Chemistry (IUPAC), the nitrogen adsorption–desorption isotherms of all four biochars are Type IV isotherms. Combined with the pore volume–pore size distribution diagrams, it can be concluded that these biochars exhibit distinct mesoporous structures (with pore sizes ranging from 2 to 50 nm). In the low relative pressure region ($P/P_0 < 0.1$), the adsorption volume rises rapidly, indicating the presence of microporous structures (with pore sizes < 2 nm) within the biochars. In the medium-pressure region ($P/P_0 = 0.3–0.9$), the rate of increase in adsorption volume slows down, accompanied by the appearance of a “hysteresis loop,” suggesting that multilayer adsorption and capillary condensation predominantly occur within the mesopores. In the high-pressure region, the hysteresis loop persists, indicating that the desorption behavior of mesopores and macropores is still influenced by capillary condensation effects. The hysteresis loops of LBC-500 (1:3) and LBC-600 (1:3) more closely resemble the H2(b) type, suggesting that their mesoporous structures are primarily ink-bottle shaped. The occurrence of this “hysteresis loop” may be related to pore blockage. In contrast, the hysteresis loops of LBC-500 and LBC-600 are smaller and tend towards the H3 type. This type of “hysteresis loop” is commonly observed in aggregates with layered structures, where the pores may predominantly consist of a mixed structure of slit-shaped mesopores and macropores. The differences in pore volume and adsorption capacity can be attributed to the varying contributions of mesopores and macropores to nitrogen adsorption.

It can be observed from Table 2 that, among the four types of biochars, LBC-500 and LBC-600 had relatively small specific surface areas. This indicates that, in this experiment, merely increasing the pyrolysis temperature was insufficient to enhance the specific surface area of lotus leaf biochar and might even lead to a reduction in it. After modification with KOH, the specific surface areas of the biochars were significantly increased. Moreover, a higher pyrolysis temperature resulted in a larger specific surface area for the KOH-modified biochars. Similarly, in terms of pore volume, LBC-600 (1:3) exhibited the highest pore volume. Regarding the average pore size, LBC-600 had the largest pore size, while LBC-600 (1:3) had the smallest. This suggests that, at the same temperature, the addition of KOH caused the biochars to preferentially generate micropores, leading to a shift towards smaller pore sizes.

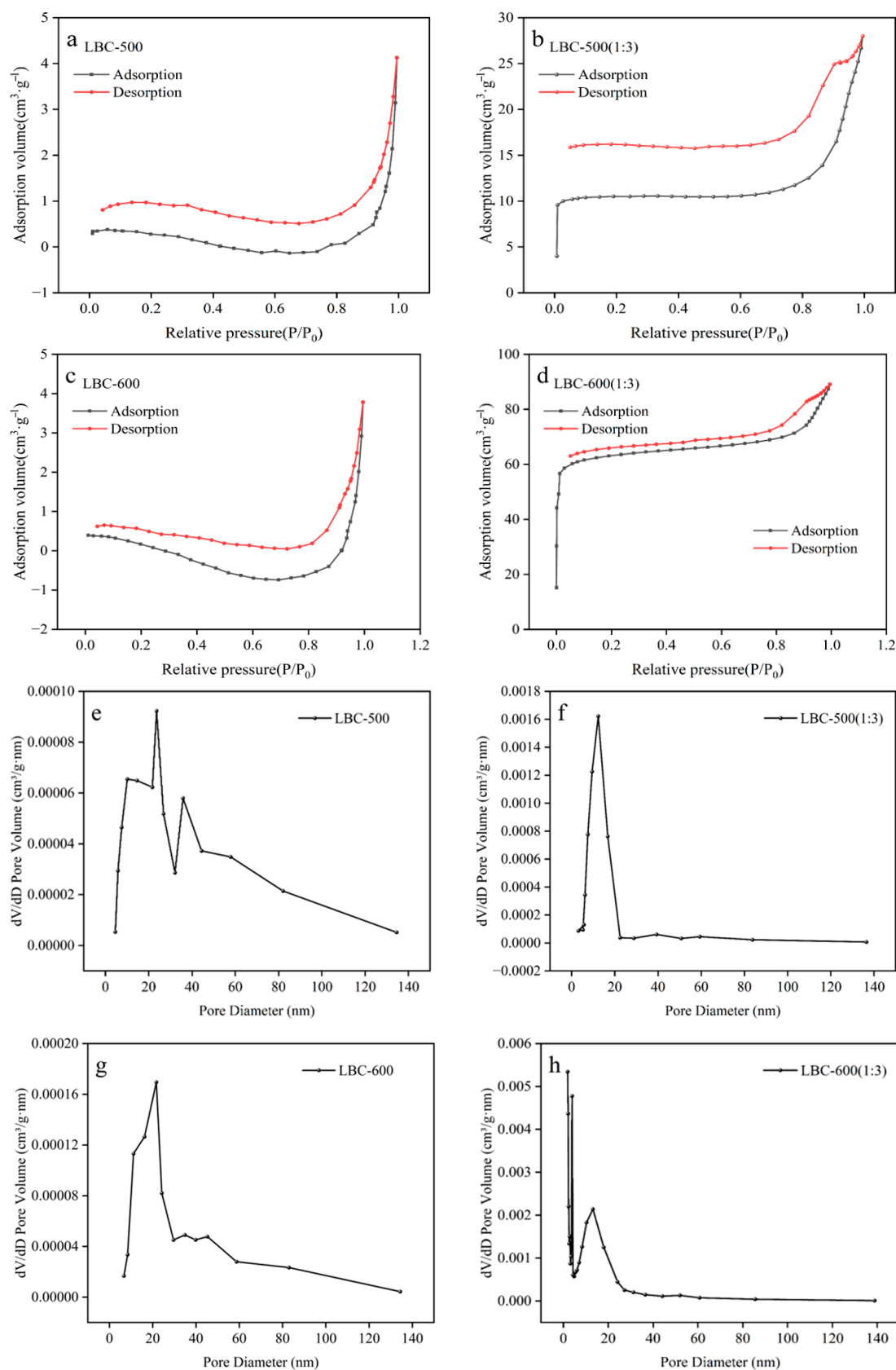


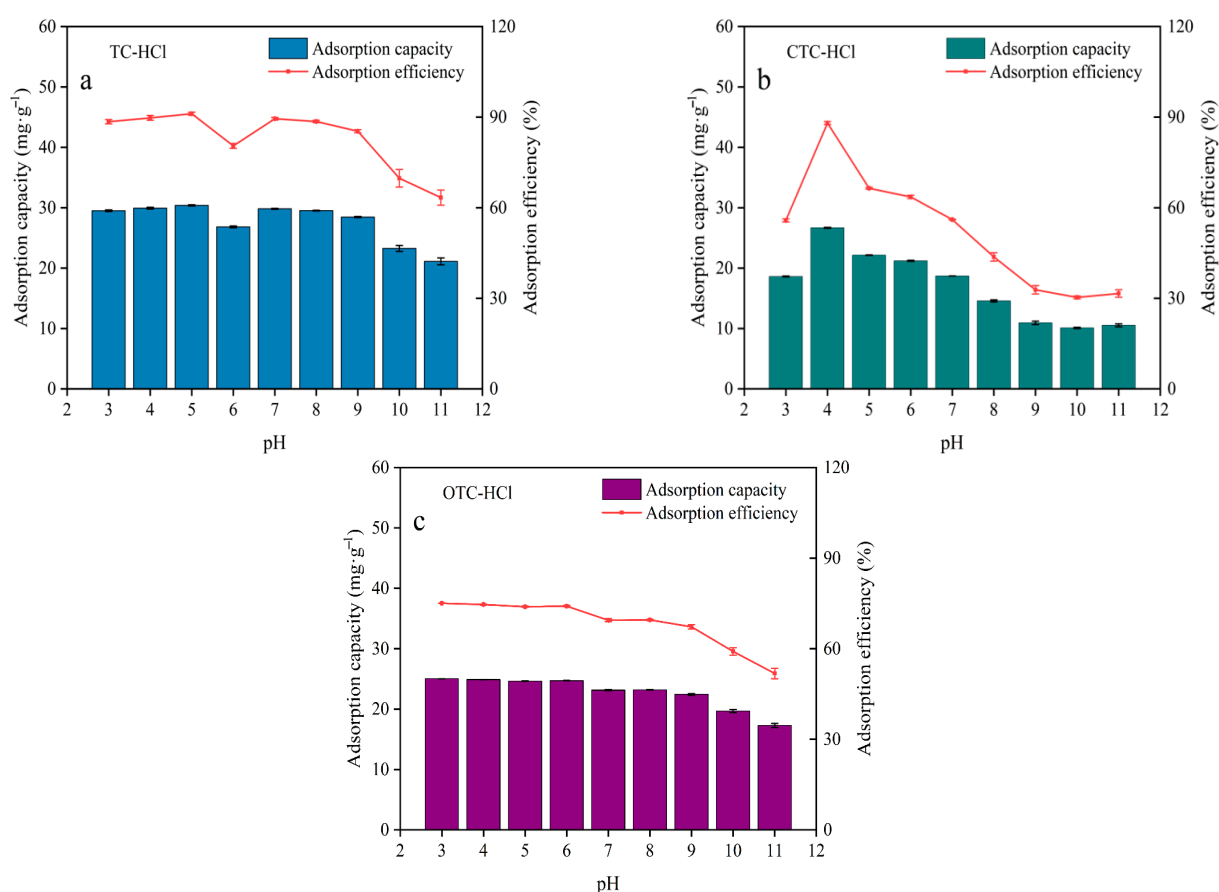
Figure 10. BET characterization of four kinds of biochar Nitrogen adsorption–desorption: (a): LBC-500, (b): LBC-500 (1:3), (c): LBC-600, (d): LBC-600 (1:3). Pore volume–aperture distribution: (e): LBC-500, (f): LBC-500 (1:3), (g): LBC-600, (h): LBC-600 (1:3).

Table 2. Table of specific surface area and pore structure parameters of four kinds of biochar.

Name	Specific Surface Area ($\text{m}^2 \cdot \text{g}^{-1}$)	Pore Volume ($\text{cm}^3 \cdot \text{g}^{-1}$)	Average Pore Diameter (nm)
LBC-500	1.58	0.0064	51.76
LBC-500 (1:3)	41.98	0.0290	24.06
LBC-600	1.49	0.0058	63.92
LBC-600 (1:3)	247.70	0.0488	19.11

3.3. Influence of pH

To quantitatively evaluate the impact of pH on the adsorption of three antibiotics (TC-HCl, CTC-HCl, and OTC-HCl) by LBC-600 (1:3), the variation trends of adsorption efficiency and adsorption capacity were measured, and the results are presented in Figure 11. It can be observed from the figure that the adsorption performance of the three antibiotics was better under acidic conditions and significantly decreased in alkaline environments, which was consistent with the previous findings [29]. Moreover, the adsorption behaviors of the three antibiotics exhibited similar trends with changes in pH. This might be attributed to the similar tetracycline parent nuclei and the arrangement of key functional groups among tetracycline, chlortetracycline, and oxytetracycline, which resulted in similar chemical behaviors (such as adsorption properties and dissociation characteristics) [30].

**Figure 11.** Effect of LBC-600 (1:3) on adsorption of antibiotics in different pH environments. (a): TC-HCl, (b): CTC-HCl, (c): OTC-HCl.

3.4. Influence of Adsorbent Dosage

Figure 12 illustrates the effect of the dosage of LBC-600 (1:3) on the adsorption of three antibiotics. As the dosage of LBC-600 (1:3) increased, the adsorption efficiency for the three antibiotics gradually rose, while the adsorption capacity gradually decreased. When

the dosage of LBC-600 (1:3) was low, the biochar provided a limited number of adsorption sites, which were effectively utilized to adsorb pollutants in the water. However, as the dosage increased, the surface of the biochar might approach saturation. With the addition of more LBC-600 (1:3), the utilization efficiency of its adsorption sites gradually declined. Although the total mass of biochar increased with the dosage, the amount of pollutants adsorbed per unit mass of biochar decreased, leading to a reduction in adsorption capacity. At low dosages, the concentration of pollutants in the solution was relatively high, and the adsorption sites on the biochar surface were limited, resulting in a lower proportion of pollutants being removed. Nevertheless, as the dosage increased, the biochar was able to adsorb more pollutants, leading to a gradual increase in adsorption efficiency.

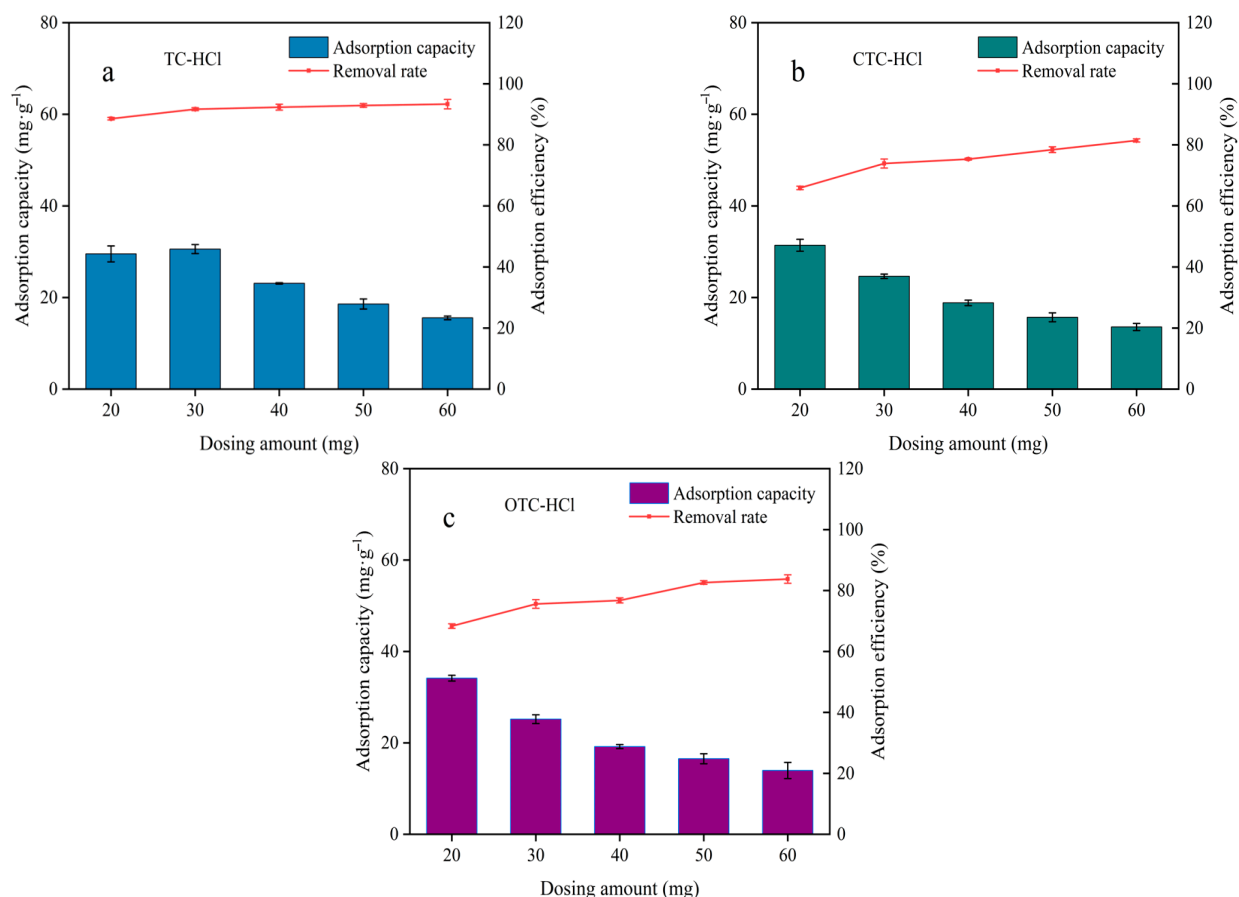


Figure 12. Effect of LBC-600 (1:3) dosage on adsorption effect of three antibiotics. (a): TC-HCl, (b): CTC-HCl, (c): OTC-HCl.

3.5. Effect of Initial Concentration of Antibiotics

Figure 13 presents the variation curves of the adsorption effect of LBC-600 (1:3) on three antibiotics with respect to their concentrations. Similarly, the adsorption efficiency increased with the initial concentration of the antibiotics. At low concentrations (20 mg·L⁻¹), LBC-600 (1:3) exhibited the highest adsorption efficiencies for the three antibiotics (TC-HCl = 91.14%, CTC-HCl = 68.04%, OTC-HCl = 57%). As the initial concentration increased, the adsorption efficiency gradually decreased, while the adsorption capacity gradually increased and then tended to reach equilibrium. The adsorption capacities of LBC-600 (1:3) for the three antibiotic molecules at various concentrations were different, with the order being QTC-HCl > QCTC-HCl > QOTC-HCl. The reason for this order of adsorption capacities was that the tetracycline molecule had a relatively small molecular weight of 444.43 Da and low steric hindrance, making it easier to diffuse into the pores of the biochar and thus

more readily adsorbed. For CTC-HCl and OTC-HCl, their molecular weights were similar, at 513.34 Da and 496.60 Da, respectively. However, the presence of a chlorine atom at the C7 position in the CTC-HCl molecule increased the electron cloud density, thereby enhancing the π - π interactions with the biochar surface. Additionally, the chlorine atom significantly improved the hydrophobicity of CTC-HCl, making it more inclined to adsorb onto the biochar surface through hydrophobic interactions [31]. In contrast, OTC-HCl had an additional hydroxyl group (-OH) at the C5 position, which significantly increased the hydrophilicity of the antibiotic molecule. Moreover, due to its limited diffusion ability, OTC-HCl had the weakest adsorption competitiveness.

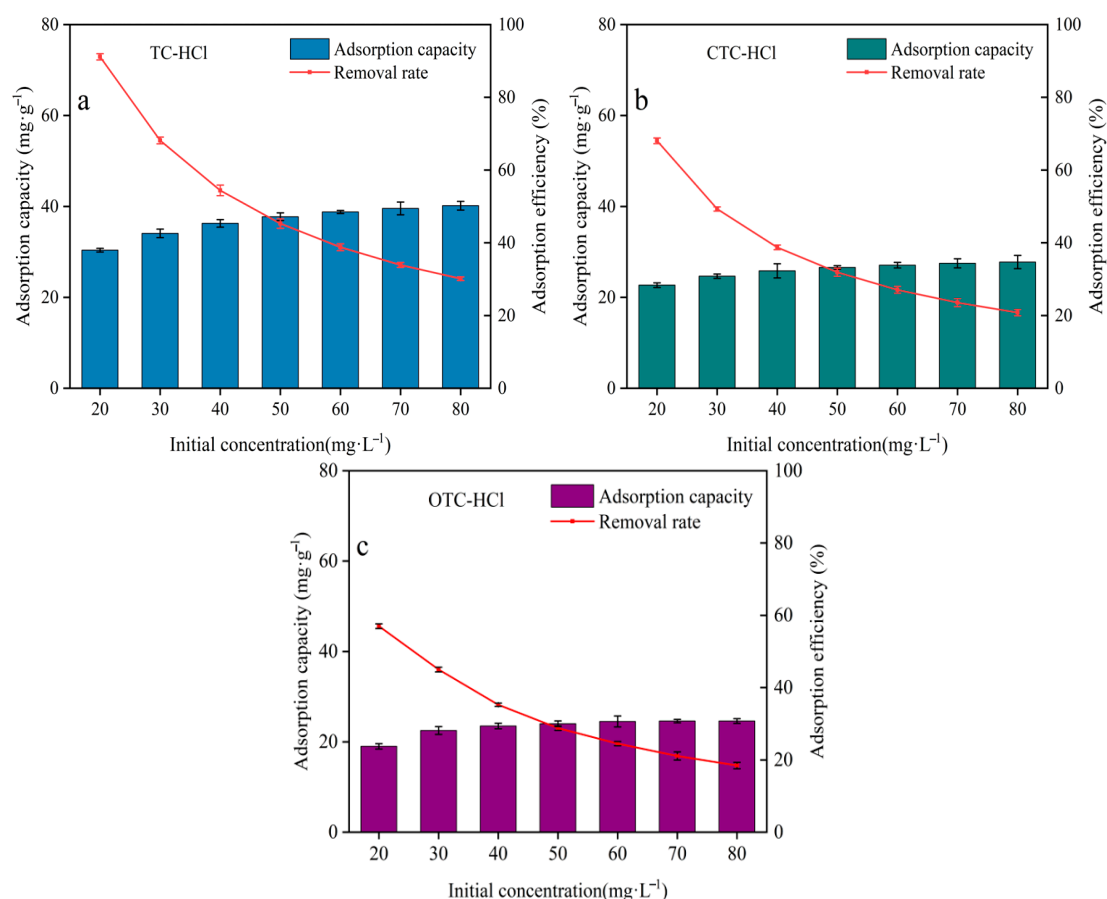


Figure 13. Effect of initial concentration on LBC-600 (1:3) adsorption of antibiotics. (a): TC-HCl, (b): CTC-HCl, (c): OTC-HCl.

3.6. Effect of Common Salts in Water

Figure 14 illustrates the effect of common salts on the adsorption of antibiotics by LBC-600 (1:3). Among the three antibiotics, CTC-HCl was most significantly inhibited by ions. OTC-HCl had a relatively low initial adsorption capacity but was less inhibited by ions compared to CTC-HCl. TC-HCl maintained a relatively high adsorption capacity even at high ionic concentrations. For TC-HCl, the degree of cationic inhibition followed the order: $\text{Na}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$, while the degree of anionic inhibition was: $\text{NO}_3^- < \text{Cl}^- < \text{SO}_4^{2-}$. For CTC-HCl, the cationic inhibition order was: $\text{Mg}^{2+} < \text{Na}^+ < \text{Ca}^{2+}$, and the anionic inhibition order was the same as that of TC-HCl: $\text{NO}_3^- < \text{Cl}^- < \text{SO}_4^{2-}$. For OTC-HCl, the cationic inhibition order was: $\text{Na}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$, and the anionic inhibition order was: $\text{Cl}^- < \text{NO}_3^- < \text{SO}_4^{2-}$. Cations, with their positive charges, could effectively screen the negative charges on the surface of LBC-600 (1:3), reducing electrostatic attraction with antibiotic molecules and competing for adsorption sites on the biochar surface. In particular,

Ca^{2+} and Mg^{2+} , due to their higher charges and hydrated ionic radii more similar to those of antibiotic molecules, exhibited stronger adsorption competition, leading to a significant decrease in the adsorption of antibiotics by the biochar. SO_4^{2-} , carrying a double negative charge, had a strong ionic strength, which greatly weakened the electrostatic adsorption of the biochar. Therefore, SO_4^{2-} had a greater interfering effect on the adsorption of antibiotics by the biochar compared to the other two anions. OTC-HCl, with an additional hydroxyl group at the C5 position, provided more binding sites for Na^+ , resulting in stronger cationic interference compared to TC-HCl. The chlorine atom at the C7 position of CTC-HCl gave it a higher electronegativity, enhancing its coordination ability with Ca^{2+} and Mg^{2+} to form more stable complexes. Thus, CTC-HCl was more significantly inhibited in the presence of cations. Meanwhile, the chlorine substituent of CTC-HCl altered the charge distribution of the antibiotic molecule, enhancing electrostatic repulsion with SO_4^{2-} [31].

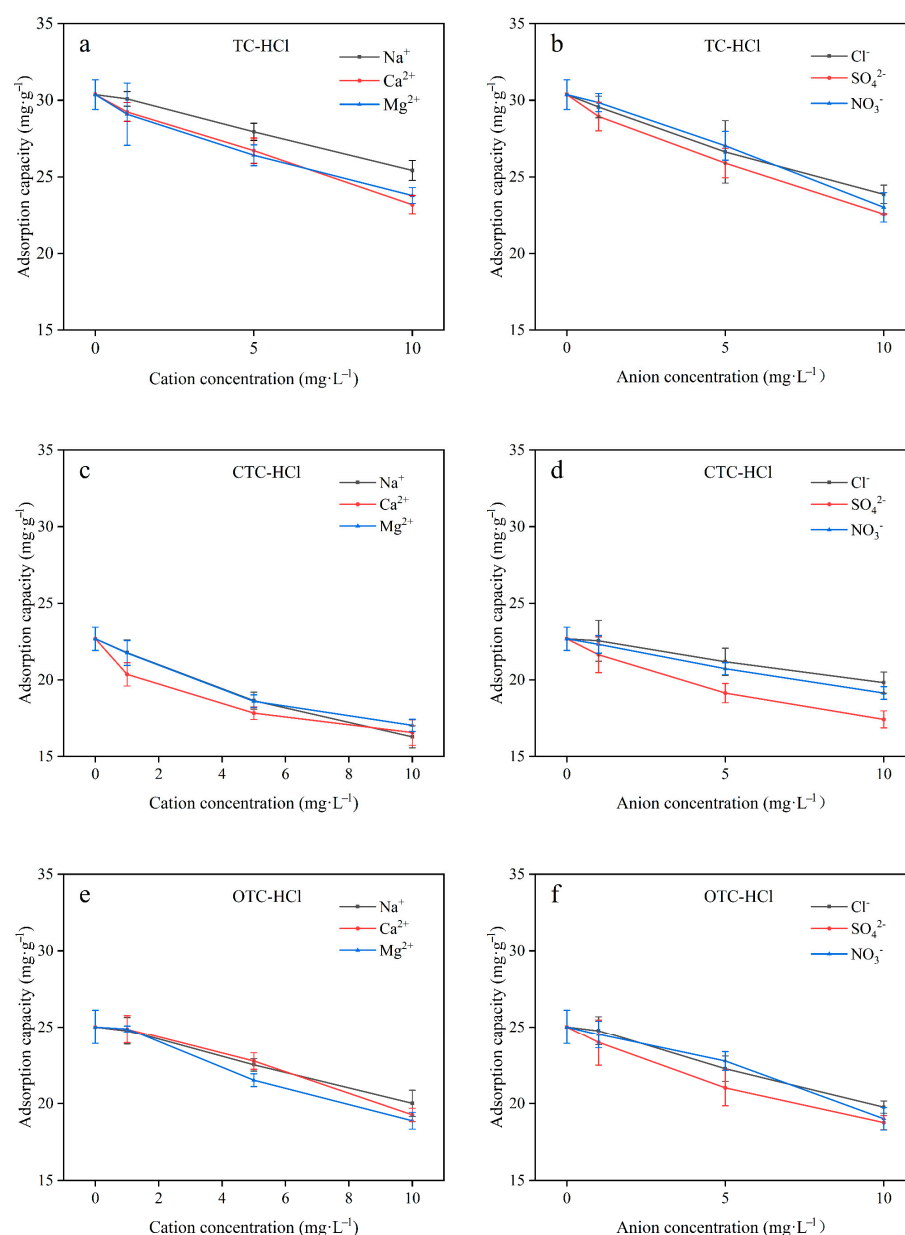


Figure 14. Effect of cationic and anion coexistence on adsorption of three antibiotics by LBC-600 (1:3). (a,c,e) are the adsorption effects of LBC-600 (1:3) on TC-HCl, CTC-HCl, and OTC-HCl when cations coexist, respectively. (b,d,f) are the adsorption effects of LBC-600 (1:3) on TC-HCl, CTC-HCl, and OTC-HCl when anions coexist, respectively.

3.7. Assessment of Adsorbent Recycling Utilization

Figure 15 presents the changes in the adsorption performance of LBC-600 (1:3) for antibiotics with respect to the number of cycles. As the number of cycles increased, the adsorption capacities of LBC-600 (1:3) for the three antibiotics all showed a decreasing trend, but the decline was relatively small, indicating that the adsorption performance of the material remained relatively stable after multiple cycles. After six cycles of adsorption, the recycling utilization rate of LBC-600 (1:3) was still above 69%. Among them, LBC-600 (1:3) exhibited the best adsorption stability for TC-HCl, with a recycling utilization rate of 81.8% after six cycles, followed by CTC-HCl (71.56%), and OTC-HCl had the lowest rate (69.8%). The possible reason for the decrease in recycling utilization rate was that, as the number of adsorption cycles increased, the relevant sites on the biochar surface were occupied by antibiotic molecules and could not be completely desorbed, leading to a reduction in the available adsorption sites.

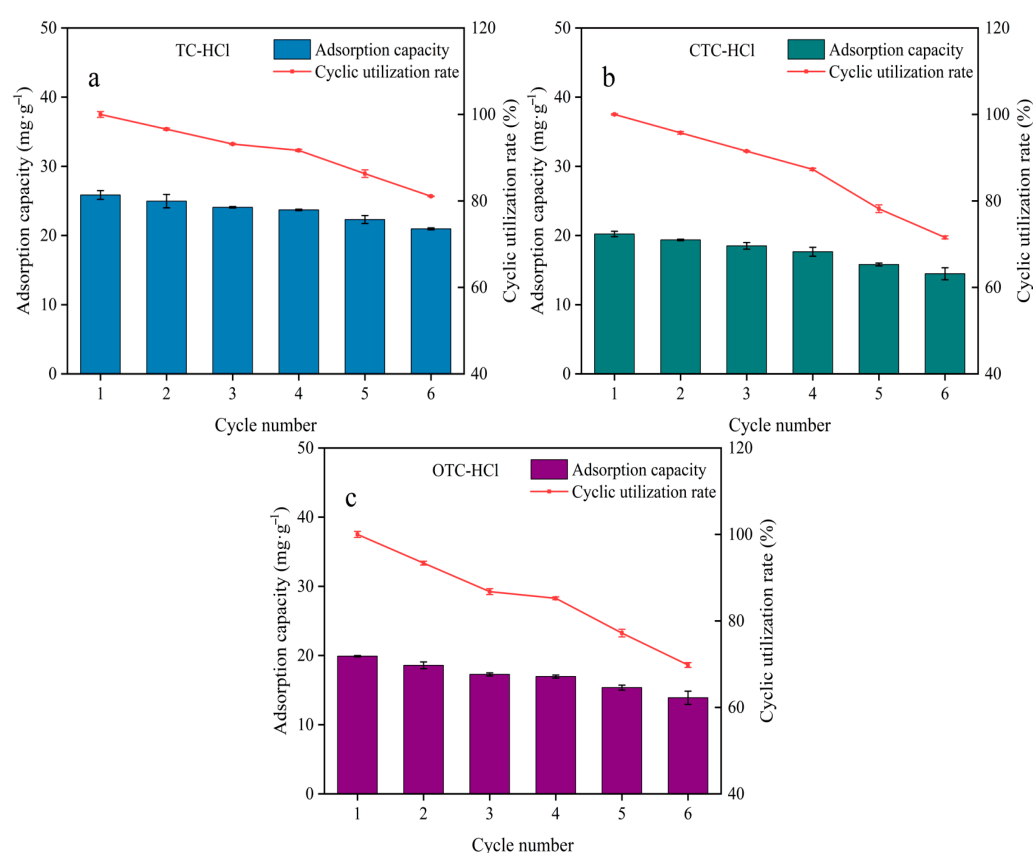


Figure 15. Effect of cyclic adsorption of LBC-600 (1:3) on the adsorption effect of antibiotics. (a): TC-HCl, (b): CTC-HCl, (c): OTC-HCl.

3.8. Adsorption Kinetics Analysis

Figure 16a–c present the variation curves of the adsorption effects of LBC-600 (1:3) on three antibiotics (TC-HCl, CTC-HCl, and OTC-HCl) over time. For these three antibiotics, the coefficient of determination (R^2) values for the pseudo-second-order kinetic model were 0.990, 0.982, and 0.989, respectively, which were all higher than those for the pseudo-first-order kinetic model (0.975, 0.972, and 0.967). This indicated that the pseudo-second-order kinetic model could more accurately describe the adsorption process of LBC-600 (1:3) for the three antibiotics. Moreover, the adsorption equilibrium quantities fitted by the pseudo-second-order kinetic model had smaller deviations compared to those fitted by the pseudo-first-order kinetic model and were closer to the actual adsorption values. It was thereby proven that the adsorption process of LBC-600 (1:3) for the three antibiotics was

mainly dominated by chemisorption (involving hydrogen bonding and π - π interactions). The poor fitting of the pseudo-first-order kinetic model suggested that physical adsorption accounted for a relatively small proportion in the overall adsorption process. As shown in Table 3, CTC-HCl had the highest K_2 value among the three antibiotics, which was attributed to its strong hydrophobicity, enabling it to be rapidly adsorbed by LBC-600 (1:3). For TC-HCl, although it had the smallest K_2 value, it had the largest Q_e value, indicating that a longer contact time might be required.

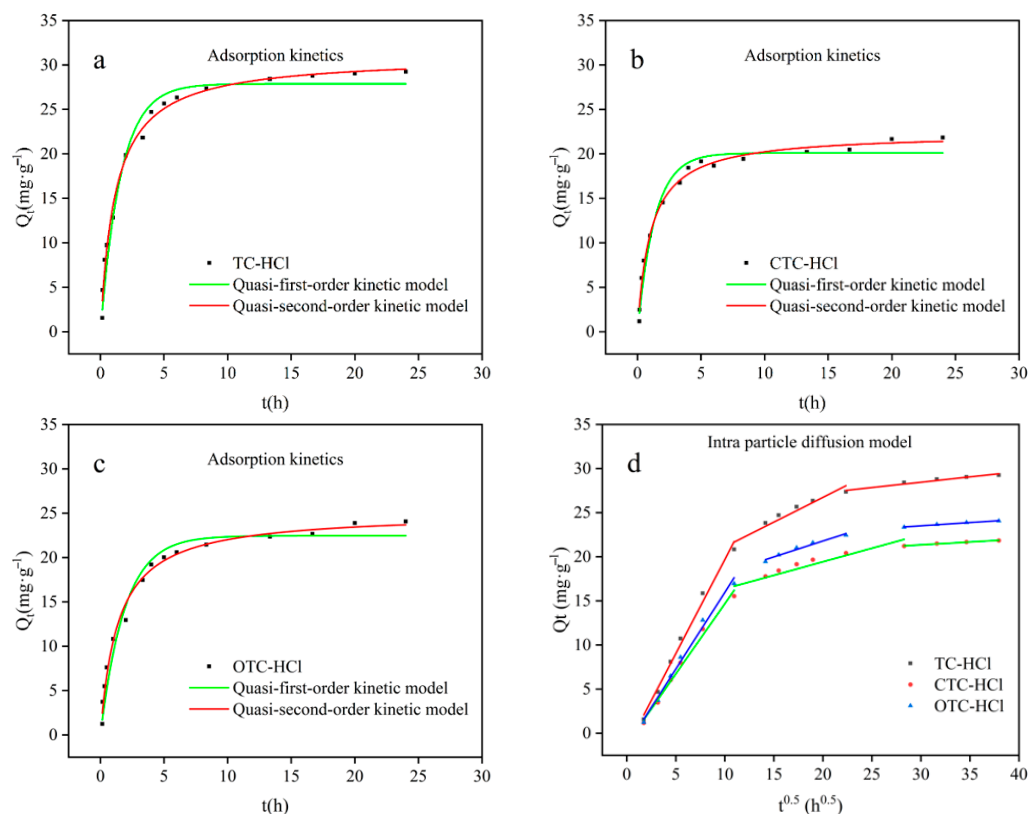


Figure 16. Adsorption kinetics models (a–c) of LBC-600 (1:3) for three antibiotics and intraparticle diffusion model. (a): TC-HCl, (b): CTC-HCl, (c): OTC-HCl, and (d): intramaterial diffusion model.

Table 3. Fitting parameter table of quasi-first-order and quasi-second-order dynamic models.

Model Adsorbate	Quasi-First-Order Dynamic Model			Quasi-Second-Order Dynamic Model		
	K_1 (h ⁻¹)	Q_e (mg·g ⁻¹)	R_2	K_2 (g·mg ⁻¹ ·h ⁻¹)	Q_e (mg·g ⁻¹)	R_2
TC-HCl	0.62	27.88	0.975	0.02	30.98	0.990
CTC-HCl	0.73	20.10	0.972	0.04	23.34	0.982
OTC-HCl	0.53	22.47	0.967	0.03	25.05	0.989

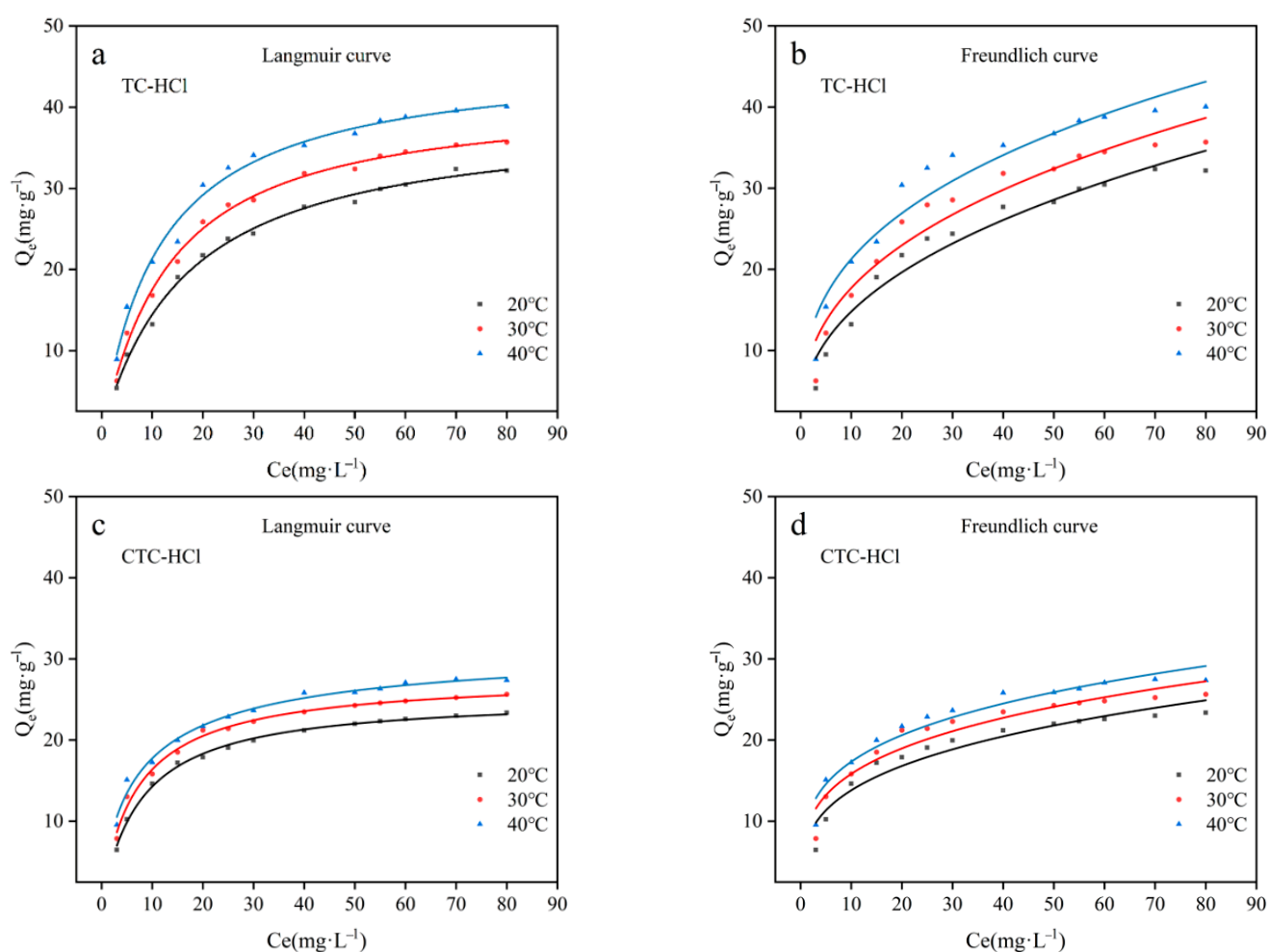
The adsorption process was analyzed using the intraparticle diffusion model. Based on Table 4 and Figure 16d, due to the excellent adsorption performance of LBC-600 (1:3), the adsorption effects of LBC-600 (1:3) on the three antibiotics showed a linear increase with time during the initial adsorption stage (0–5 h). The slope was the largest in this first stage, indicating that the three antibiotics migrated from the solution to the external surface of the biochar [32]. During the intermediate adsorption stage (5–15 h), as the adsorption sites on the biochar surface gradually became saturated, intraparticle diffusion became the main limiting factor. In the later adsorption stage (15–24 h), the adsorption process tended to reach equilibrium.

Table 4. Table of fitting parameters in different stages of intraparticle diffusion model.

	Phase 1			Phase 2			Phase 3		
	k_{i1}	C_{i1}	R_2	K_{i2}	C_{i2}	R_2	K_{i3}	C_{i3}	R^2
TC-HCl	2.13	−1.60	0.99	0.56	15.54	0.91	0.12	24.83	0.94
CTC-HCl	1.59	−1.21	0.96	0.31	13.31	0.85	0.06	19.42	0.98
OTC-HCl	1.74	−1.41	0.98	0.36	14.61	0.96	0.07	21.31	0.97

3.9. Adsorption Isotherm Experiments

Figure 17 displays the adsorption isotherms of TC-HCl, CTC-HCl, and OTC-HCl by LBC-600 (1:3). According to the fitting parameter table (Table 5), it can be observed that both the Langmuir model and the Freundlich model can adequately fit the adsorption effects of LBC-600 (1:3) on the three antibiotics. The Langmuir model, with higher R^2 values, particularly demonstrated superior fitting capability of LBC-600 (1:3) for the three TCs. Thus, LBC-600 (1:3) predominantly acts as a heterogeneous adsorbent, facilitating monolayer adsorption of these three TCs. The $1/n$ of the Freundlich equation for the TC adsorption by LBC-600 (1:3) was below 1, indicating favorable adsorption conditions. Based on the Langmuir model, the maximum adsorption capacities of LBC-600 (1:3) for TC-HCl, CTC-HCl, and OTC-HCl were 46.2, 31.0, and 32.3 $\text{mg}\cdot\text{g}^{-1}$ at 40 °C, respectively. The adsorption capacity of LBC-600 (1:3) for TC-HCl was much higher than that for CTC-HCl and OTC-HCl.

**Figure 17.** Cont.

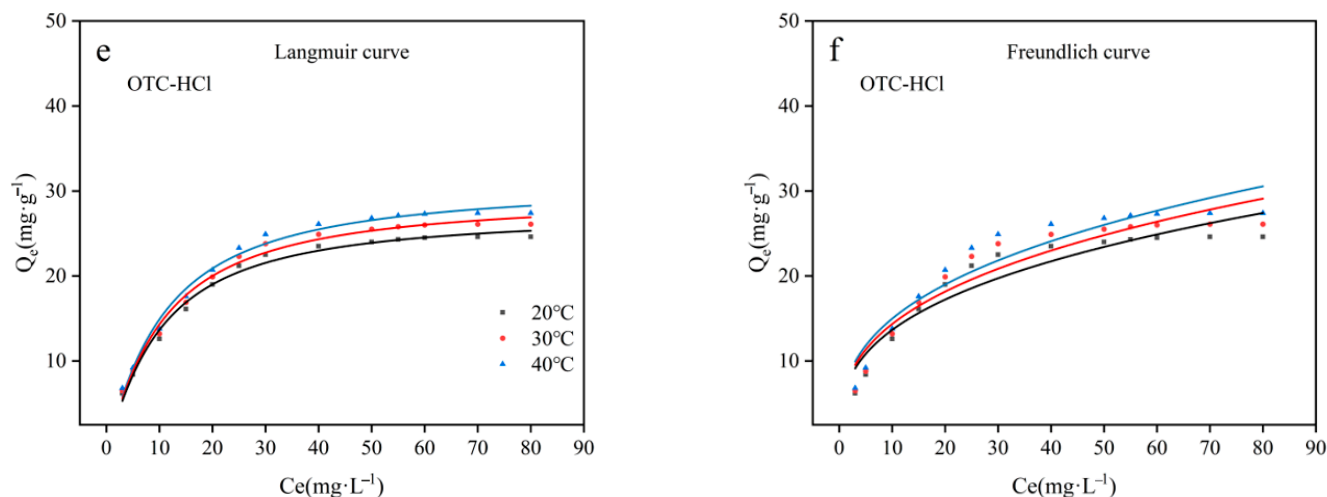


Figure 17. Adsorption isothermal experiment of LBC-600 (1:3) for three antibiotics. Langmuir curve: (a): LBC-600 (1:3) adsorbed TC-HCl, (c): LBC-600 (1:3) adsorbed CTC-HCl, (e): LBC-600 (1:3) adsorbed OTC-HCl. Freundlich curve: (b): LBC-600 (1:3) adsorbed TC-HCl, (d): LBC-600 (1:3) adsorbed CTC-HCl, (f): LBC-600 (1:3) adsorbed OTC-HCl.

Table 5. Table of adsorption isotherm fitting parameters for Langmuir and Freundlich models.

Model Temperature		Langmuir			Freundlich		
		Q_m (mg·g ⁻¹)	K_a	R^2	K_f	$1/n$	R^2
TC-HCl	20 °C	37.7	0.0562	0.993	5.77	0.409	0.952
	30 °C	41.1	0.0635	0.994	7.45	0.375	0.936
	40 °C	46.2	0.0868	0.987	9.71	0.340	0.930
CTC-HCl	20 °C	27.5	0.0642	0.988	6.31	0.335	0.897
	30 °C	29.4	0.0645	0.990	6.56	0.340	0.901
	40 °C	31.0	0.0651	0.993	6.80	0.342	0.904
OTC-HCl	20 °C	25.4	0.125	0.995	7.19	0.283	0.920
	30 °C	28.0	0.156	0.989	8.68	0.261	0.915
	40 °C	32.3	0.208	0.984	9.75	0.249	0.939

3.10. Comparison of Adsorption Effect with Other Biochar Materials

As can be seen from Table 6, there were significant differences in the adsorption capacities of different biochars for tetracycline-class antibiotics. The LBC-600 (1:3) prepared in this experiment exhibited a relatively high adsorption capacity, indicating that its adsorption performance was comparable to, if not better than, that of other biochars. Furthermore, from the study on the effect of initial concentration on adsorption efficiency, it was observed that, even at a high concentration (50 mg·L⁻¹), the adsorption efficiency of LBC-600 (1:3) for antibiotics remained higher than that of cow-dung-based biochar, with a corresponding biochar dosage (0.6 g·L⁻¹) much lower than the dosage of cow dung biochar (1.25 g·L⁻¹). Therefore, it could be inferred that the biochar prepared in this experiment possessed relatively stable concentration adaptability.

LBC-600 (1:3) demonstrated varying adsorption selectivities for different tetracycline-class antibiotics, showing an adsorption trend of TC-HCl > OTC-HCl > CTC-HCl. This might limit its application in the treatment of mixed antibiotic wastewater. Meanwhile, compared to sludge-based biochar, LBC-600 (1:3) had a lower adsorption capacity. Thus, it might have certain limitations when treating high-concentration wastewater (≥50 mg·L⁻¹).

Table 6. Adsorption capacity of different biochars for tetracycline antibiotics.

Biochar Precursor	Preparation Method	Charcoal Dosage	Initial Concentration (mg·L ⁻¹)	Adsorption Capacity (mg·g ⁻¹)	Reference
Cow dung	700 °C pyrolysis	1.25 g·L ⁻¹	50	11.79	[32]
Sludge	600 °C, KOH modified	25 mg	100	154.16	[33]
Straw	600 °C, 30%(w/w) Ca(OH) ₂ modified	0.1 g	50	40	[34]
Xianggu	700 °C pyrolysis	1.6 g·L ⁻¹	10	17.68	[27]
Lotus leaf	Preparation by pyrolysis at 600 °C and modification with KOH mass ratio of 1:3	30 mg	20	29.26 (TC-HCl)	This study
				21.84 (CTC-HCl)	
				24.07 (OTC-HCl)	

4. Conclusions

This study aims to utilize abundant wetland plant residues (lotus leaves, giant reeds, and canna lilies) for biochar production and employ the prepared biochar for common antibiotic (TC) removal in water environment via an adsorption process. Recycling of wetland plant residues and the removal of tetracycline antibiotics from water environments can be simultaneously achieved. The main conclusions are as follows:

(1) Among the three wetland plants, lotus leaf biochar exhibited superior adsorption performance under different KOH ratios and temperatures. When the KOH ratio was 1:3, its adsorption efficiency was significantly enhanced, reaching 71.68%, 65.16%, and 67.43% for TC-HCl, CTC-HCl, and OTC-HCl, respectively. As the temperature increased, the adsorption efficiency of lotus leaf biochar gradually improved.

(2) Biochars prepared from plant residues through high-temperature pyrolysis and KOH modification could significantly enhance the adsorption efficiency for target pollutants. Lotus leaf biochar, which underwent pyrolysis at 600 °C and modification with a plant residue to KOH mass ratio of 1:3, achieved high adsorption capacities for TC-HCl, CTC-HCl, and OTC-HCl, with values of 30.56, 24.63, and 25.19 mg·g⁻¹, respectively. A possible reason for this phenomenon was that the surface area of biochar increased with the rise in temperature, which consequently enhanced its adsorption capacity for antibiotics.

(3) Acidic conditions were more favorable for the adsorption of the three antibiotics by biochars. The addition of other cations and anions to the solution could produce certain interference with the adsorption performance of biochars. The higher the concentration, the stronger the interference effect. As the number of adsorption cycles increased, the relevant sites on the surface of biochar could not be completely desorbed after being occupied by antibiotic molecules. This led to a reduction in the available adsorption sites. Consequently, the adsorption capacity of biochar significantly decreased after six adsorption cycles. However, the recycling rate of biochar remained relatively high.

(4) The fitting results of the adsorption isotherms showed that the adsorption processes of the three antibiotics on LBC-600 (1:3) conformed to the Langmuir model. Therefore, it was indicated that the surface of LBC-600 (1:3) had uniform adsorption sites, and the adsorption process was completed through chemisorption on a monolayer. The straight line in the first stage of the intraparticle diffusion model did not pass through the origin, which suggested that intraparticle diffusion was not the sole rate-limiting step in the adsorption process. Oxygen-containing functional groups participated in the adsorption through π - π interactions, hydrogen bonding, electrostatic attraction, surface complexation, and pore filling.

(5) Antibiotics in water environment are mainly derived from real wastewaters, which may contain many components such as some organic matters and salts. Thus, studies on antibiotic removal from real wastewaters should be conducted in further works, which can enhance the application potential of biochar in real projects.

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